Tony Siu

Founding AI Engineer · Multi Modal GenAI researcher · Applied Statistics Researcher

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Professional Experience

Founding AI Engineer
San Francisco, CA

A16z Speedrun SR005 — *Apr 2025 - Dec 2025*

- Architected & built a Natural-language-to-full GenAI game generation MVP for games (engine, assets, logic) in under 30 days
- Principal Engineer that delivered ~60% of core engineering output of a 20-person startup, while owning architecture, execution, and delivery
- 200× performance optimization to reduce sprite/GIF rendering from 5s per animation to 0.2s across 78+ concurrent assets
- Developed vision-based, promptless agentic loops for long-context, cross-session user memory and task automation
- Reimplemented bleeding-edge research (ACE, Tongyi, AgentFold papers) into production-grade CUA scaffolds and production agent runtimes

Code&Coffee Philadelphia Chapter Founder
Philadelphia, PA

Code&Coffee — *Jan 2024 - Present*

- Founded and led Philadelphia's largest active, builder-focused nonprofit tech community, scaling from 2 founders to 3,200+ members in one year
- Built a cross-domain ecosystem connecting engineers, founders, researchers, VCs, and mentors across AI, systems, and product
- Produced high-impact programming including Philly Tech Week 2025's only builder-focused hackathon, founder demo days, and technical meetups (AI papers, coworking, LeetCode, coffee-and-code)
- How Tony Siu built Philly's 3k+ member Code and Coffee meetup. *Technical.ly*, 2026.
- Code & Coffee Philadelphia's Mission to Revive Community-Driven Learning. *TechTimes*, 2026.
- How Research Projects & Community Work Shape His Approach to Building. *IBTimes*, 2026.

Multi-Modal Generative AI Researcher
Philadelphia, PA

Temple College of Science & Technology — *May 2023 - May 2025*

- Authored multiple papers submitted to IEEE, CAIP, and ECAI on multimodal computer vision and video understanding
- Improved performance on video QA performance by 6% using rPPG features from RhythmFormer and RhythmMamba over baseline fine-tuning
- Achieved 6% in- and out-of-sample generalization gains through Direct Preference Optimization without RLHF
- Delivered up to 9% performance improvements across benchmarks using preference alignment methods (KTO, IPO, RSO, cDPO, SSPO, GRPO)
- Generated synthetic preferred/rejected response pairs to enable scalable preference optimization without human annotation
- Evaluated SOTA video–language models (MiniGPT-4-Video, LLaVA-Next, VideoMamba) across DAiSEE, EngageNet, and SED datasets
- Reached 97% F1 on student engagement detection via engagement-level prompt fine-tuning for visual question answering

Google UR Computer Vision Research Lead
Taipei, Taiwan

Google Research — *Jan 2022 - May 2023*

- Led computer vision research for improving Google Pixel automated UI defect detection pipelines
- Secured continued Google Level-2 sponsorship through novel multimodal computer vision research outcomes
- Boosted UI flicker detection from 43% to 98% precision–recall performance using a CNN–Transformer hybrid architecture
- Expanded training data 10× via targeted augmentation, scaling from 180 to 1,991 labeled video snippets
- Reduced model training time by 20× through informed weight initialization, cutting convergence from 1,000 to 50 epochs

AI Specialist Engineer
Taipei, Taiwan

Taiwan Tech Arena — *May 2020 - Oct 2020*

- Achieved real-time 360° video stitching on NVIDIA Jetson Xavier, reducing dual 4K fisheye (170°) composePanorama latency from 6s to 0.2s
- Optimized OpenCV stitching pipeline by CUDA-integrating OpenCV 2.4 image blending into OpenCV 4.5, reducing frame time from 0.2s to 0.01s
- Enabled real-time stitching via single-pass EstimateTransform, multithreading, and stationary-camera assumptions for continuous 360° capture
- Delivered 6× faster OpenCV CUDA/cuDNN builds by packaging a single optimized compilation as reusable CPack Debian binaries
- Implemented real-time 3D-to-2D homography projections to map player field positions from panoramic video feeds
- Developed real-time sports video analytics including camera panning, zoom detection, and slow-motion event analysis

Education

Masters, Computational Data Science | GPA 4.0
Temple University, Philadelphia, PA
Jan 2024 - Current

Bachelors, Computational Data Science
Temple University, Philadelphia, PA
Aug 2022 - Dec 2024

Publications

Generalizable Detection of Student Engagement in Online Learning Environments. CAIP 2025.

rPPG Signal Enhances Student Engagement Recognition in Online Video Education. *IEEE*, 2025.